

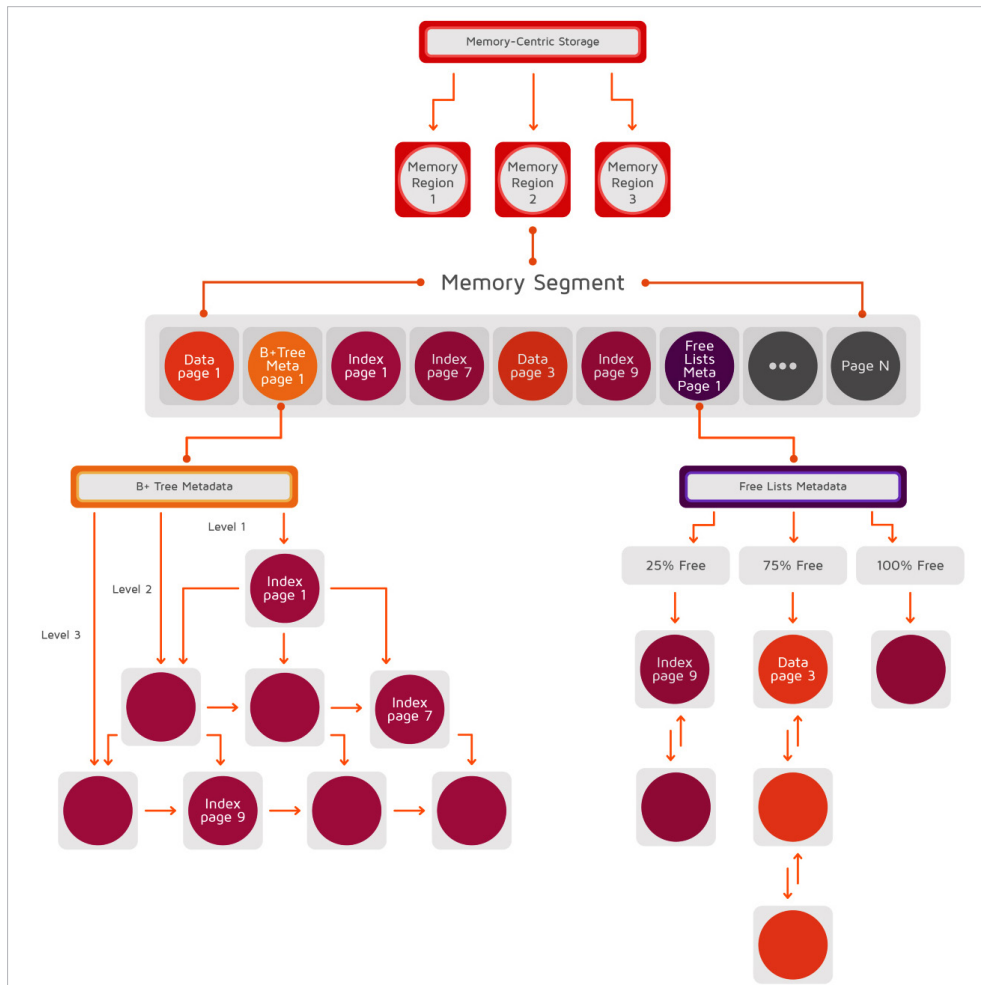


Figure 1: Memory architecture

- **Is Ignite a distributed cache?**
Yes. If Ignite native persistence is disabled, it can function as a distributed in-memory cache which assists distributed ACID transactions.
- **Is Ignite an SQL database?**
Not entirely. It assists SQL natively, although there are some differences in how Ignite handles constraints and indexes. It also supports primary and secondary indexes, but the uniqueness can only be enforced for the primary indexes. Apache Ignite does not currently support foreign key constraints.
- **Is Ignite a key-value store?**
Yes. Ignite offers a feature rich key-

value API that is JCache (JSR-107) compliant and supports Java, C++, and other languages.

- **Is Ignite a NoSQL database?**
Not precisely. Just like any other NoSQL database, Ignite is highly available and horizontally scalable. Though, unlike other NoSQL databases, it supports SQL and ACID transactions over multiple cluster nodes.
- **Is Ignite a distributed database?**
Yes. It is a distributed database with high-performance computing and in-memory speed. As we already

know, Ignite stores data as in-memory and/or on-disk and is either partitioned or replicated over a cluster of multiple nodes. This offers high performance, resiliency, and scalability.

Apache Ignite use cases

Ideal conditions for using Apache Ignite are:

- Enhancing scalability and speed of live applications
- Creating new, modern, highly performant and scalable transactional and/or analytical applications
- Building streaming analytics applications, often with Apache Kafka, Apache Spark and other streaming technologies
- Incorporating continuous deep learning and machine learning into applications to advance decision automation

If any one of these conditions is met, then

using Apache Ignite will provide significant benefits.

Use of Apache Ignite and Grid Gain is increasing, primarily as shared data and processing infrastructure across projects, to deliver in-memory speed and unlimited scalability for transactions and analytics. Apache Ignite offers a great solution to the problem of handling Big Data at very high speeds with great reliability. **END** 🐧

Reference

- <https://ignite.apache.org/>

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